

12.47

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October 4,2025

Question

The following simultaneous equations

$$x + y + z = 3$$

$$x + 2y + 3z = 4$$

$$x + 4y + kz = 6$$

will NOT have a unique solution for k equal to

- 1 0
- 2 5
- 3 6
- 4 7

Theoretical Solution

The equations can be written as,

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & k \end{pmatrix} \mathbf{x} = \begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix} \quad (1)$$

where

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Theoretical Solution

The augmented matrix is given by,

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 4 \\ 1 & 4 & k & 6 \end{array} \right) \quad (2)$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1 \implies \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 3 & k-1 & 3 \end{array} \right) \quad (3)$$

$$R_3 \rightarrow R_3 - 3R_2 \implies \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & k-7 & 0 \end{array} \right) \quad (4)$$

Theoretical Solution

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & k-7 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} \quad (5)$$

For the system to **NOT** have a unique solution, the coefficient of z in the final row must be zero.

$$k - 7 = 0 \quad (6)$$

$$\implies k = 7 \quad (7)$$

Intersection of Three Planes when $k=7$

